
CHAPTER 1

MACHINE LEARNING BASICS

NOTE: The following chapters closely follows the textbook by Peter/Norvig and the previous slides of CSE422.

1.1 Introduction

Machine learning is a subfield of artificial intelligence (AI) that focuses on developing algorithms and models that enable computers to learn patterns from data and make decisions or predictions without being explicitly programmed for specific tasks.

In simple terms, machine learning allows systems to improve their performance over time by using past experiences (i.e., data) to generalize and adapt to new situations.

1.2 Why do we need machine learning

One may ask why we need machine learning, rather than simply programming machines to perform specific tasks.

- **Unpredictable situations** For some tasks it is impossible to know all possible scenarios beforehand.

Machine learning handles unpredictable situations by learning patterns from past data and making generalizations. Even if it has not seen the exact situation before, it can make a best guess based on similar examples.

Example: Email Spam Detection

- **Training data:** Emails labeled as "spam" or "not spam".
- **New email:** The system has never seen this exact message, but it recognizes similar keywords and sender behavior from past spam.
- **Action:** Marks it as spam.

Example: Stock Investment

- **Training data:** Historical stock prices, Market reactions to past political/economic crises, Sector-wise performance during volatility.
 - **New situation:** A sudden political crisis causes uncertainty in the market which is never seen before.
 - **Action** Machine learning can help predict likely drop or recovery patterns, identify safer investment sectors and can recommend adjustment of portfolio based on learned behavior from similar past events.
- **Unknown Solutions** Some tasks are easy for humans but very hard to explicitly program using traditional rules. These tasks often involve complex patterns, uncertainty, or high variability—things that machine learning handles well by learning from data rather than following fixed instructions.

Example: Face Recognition

- **Hard to program:** Human faces vary by angle, lighting, age, expression. Writing rules for all possibilities is nearly impossible.
- **Machine learning approach:** Trains on thousands of face images, learning features (like eyes, nose shape) automatically.

Example: Handwriting Recognition

- **Hard to program:** Every person writes differently, even the same letter can look very different.
- **Machine learning approach:** Learns from many handwritten samples to recognize letters based on strokes and curves.

1.3 Paradigms of Machine Learning

Machine learning includes several paradigms that guide how systems learn from data.

Rote learning is simple memorization, like a model storing exact inputs and outputs—for example, a chatbot recalling predefined replies.

Induction involves generalizing from examples; a spam filter learns patterns in labeled emails to classify new ones.

Clustering is an unsupervised method that groups similar data, such as organizing customers into segments based on purchase behavior.

Analogy and discovery use representation mapping—like a system recognizing that the relationship between Earth and Moon is similar to Jupiter and its moons.

Genetic algorithms apply evolutionary search, creating and evolving solutions over generations, useful in optimizing investment strategies, parameter tuning etc.

Finally, **reinforcement learning** is reward-based; for instance, a game-playing agent learns to win by receiving points and improving its moves over time.

1.3.1 Major Paradigms of Machine Learning

Based on learning feedbacks we can define three major paradigms of machine learning.

- **Supervised learning:** Uses labeled data to train models, where each input has a known output. For example, a model trained to detect spam emails based on labeled examples ("spam" or "not spam").
- **Unsupervised learning:** Works with data that has no labels, helping to find hidden patterns—like clustering customers based on buying behavior without knowing their categories.
- **Reinforcement learning:** Involves an agent learning by interacting with an environment and receiving rewards or penalties; for instance, a robot learning to walk by trial and error.

1.4 Supervised Learning

Supervised learning is a type of machine learning where a model is trained on a labeled dataset—that is, each input has a corresponding correct output. The goal is for the model to learn a mapping from inputs to outputs, so it can accurately predict outcomes for new, unseen data.

In supervised learning, the model learns from **input-output pairs**, also known as **feature-label pairs**.

- **Input (Features):** These are the measurable variables that form the input vector, often denoted by X .
 - *Example:* For predicting house prices, features may include:
 - * Size of the house (in sq. ft)
 - * Number of bedrooms
 - * Location rating
 - * Age of the house
- **Output (Label or Target):** This is the value or category to be predicted, usually denoted by Y .

- *Example:* The price of the house in dollars.

Each training example is represented as a pair: (X, Y) where X is the input vector containing features $x_1, x_2, \dots$ and Y is the corresponding output label.

1.5 Steps of Supervised Learning

Supervised learning involves learning a function that maps inputs to known outputs. The general workflow includes the following steps:

1. **Collect Data:** Gather labeled data where each input has a corresponding correct output.
2. **Preprocess Data:** Clean the data, handle missing values, normalize features, and encode categorical variables if necessary.
3. **Split the Data:** Divide the dataset into training and test sets (optionally a validation set).
4. **Train the Model:** Use the training set to teach the model to learn the mapping from input to output.
5. **Evaluate the Model:** Use the test (or validation) set to measure performance using appropriate metrics.

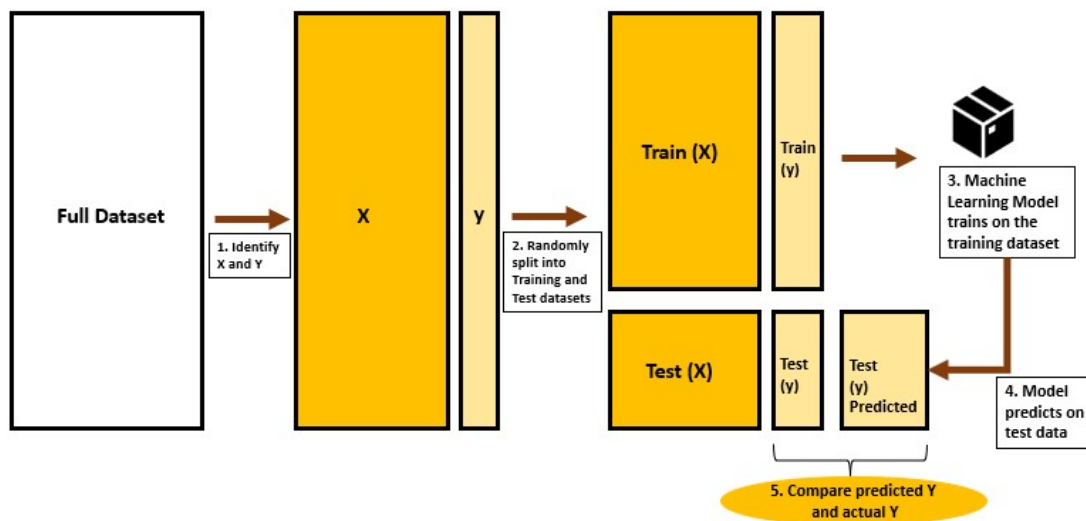


Figure 1.1: Training and evaluation of model

6. **Tune Hyperparameters:** Adjust the model settings to optimize performance.
7. **Deploy the Model:** Use the trained model to make predictions on new, unseen data.

1.6 Types of Supervised Learning

Supervised learning is mainly divided into two types:

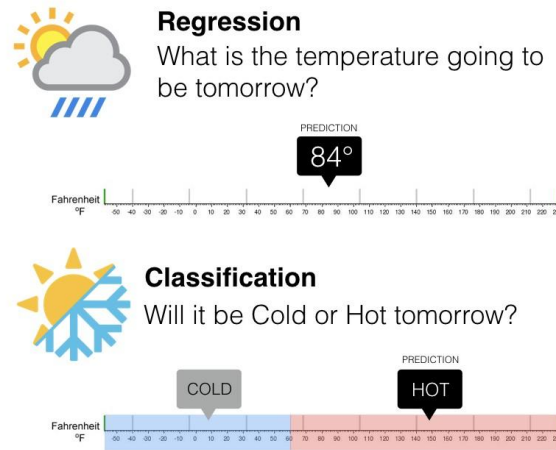


Figure 1.2: Types of supervised learning

- **Regression:**

- *Goal:* Predict continuous numeric values.
- *Example:* Predicting house prices based on size and location.

- **Classification:**

- *Goal:* Predict discrete class labels.
- *Example:* Classifying emails as “spam” or “not spam”.

1.7 Model Selection

Choosing the right model is crucial for achieving good performance. The model selection process involves:

1. **Understanding the Task:** Determine whether the problem is regression or classification.
2. **Choosing Candidate Models:** Based on task complexity, data size, and interpretability. Common models include:

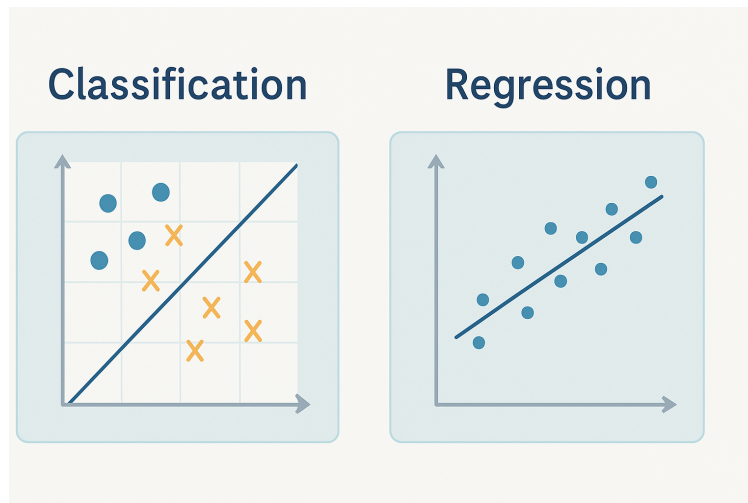


Figure 1.3: Classification Vs. Regression

- Linear/Logistic Regression
 - Decision Trees
 - k-Nearest Neighbors (k-NN)
 - Support Vector Machines (SVM)
 - Neural Networks
3. **Cross-Validation:** Evaluate models using validation techniques to estimate performance on unseen data.
 4. **Model Comparison:** Compare models based on accuracy, error rates, computation time, etc.
 5. **Select the Best Model:** Choose the model that balances performance and complexity.

1.8 Hypothesis in Supervised Learning

A **hypothesis** in supervised learning is a candidate function that maps inputs to outputs:

$$h : X \rightarrow Y$$

The goal is to find the best hypothesis h from a set called the **hypothesis space** $\mathcal{H}$, which includes all models that the learning algorithm can select from.

Example: In linear regression, a possible hypothesis could be:

$$h(x) = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

where $w_1, w_2, \dots, w_n$ are weights, and b is the bias term.

1.9 Hypothesis Selection

To select the best hypothesis from $\mathcal{H}$, the following steps are followed:

1. **Define a loss function** – Measures the error between predicted and true outputs.
 - Example: Mean Squared Error (MSE) for regression, Cross-Entropy for classification.
2. **Train the model** – Minimize the loss function on training data using optimization techniques (e.g., gradient descent).
3. **Validate** – Use a separate validation set to test the generalization ability of the hypothesis.
4. **Select** – Choose the hypothesis with the best performance on the validation set.
5. **Test** – Evaluate the final hypothesis on unseen test data to estimate real-world performance.

Supervised learning aims to find the best hypothesis that accurately maps inputs to outputs by learning from labeled data.

Bias and Variance in Hypothesis Selection

When we select a hypothesis in machine learning, we aim for a model that generalizes well to unseen data. Two important sources of error are:

Bias

- Bias is the error due to overly simplistic assumptions in the model.
- A high-bias model may under-represent the data structure.
- Example: Using a linear model on highly nonlinear data.

Variance

- Variance is the error from excessive sensitivity to small fluctuations in the training data.
- A high-variance model captures noise along with the signal.
- Example: A high-degree polynomial fitting every point exactly.

Underfitting and Overfitting

Underfitting

- Caused by a model that is too simple to capture the underlying patterns (high bias). Example: 1.4 subplot a.
- Poor performance on both training and test data.

Overfitting

- Caused by a model that is too complex and fits the noise in the training data (high variance). Example: 1.4 subplot c and subplot d.
- Excellent performance on training data but poor generalization to new data.

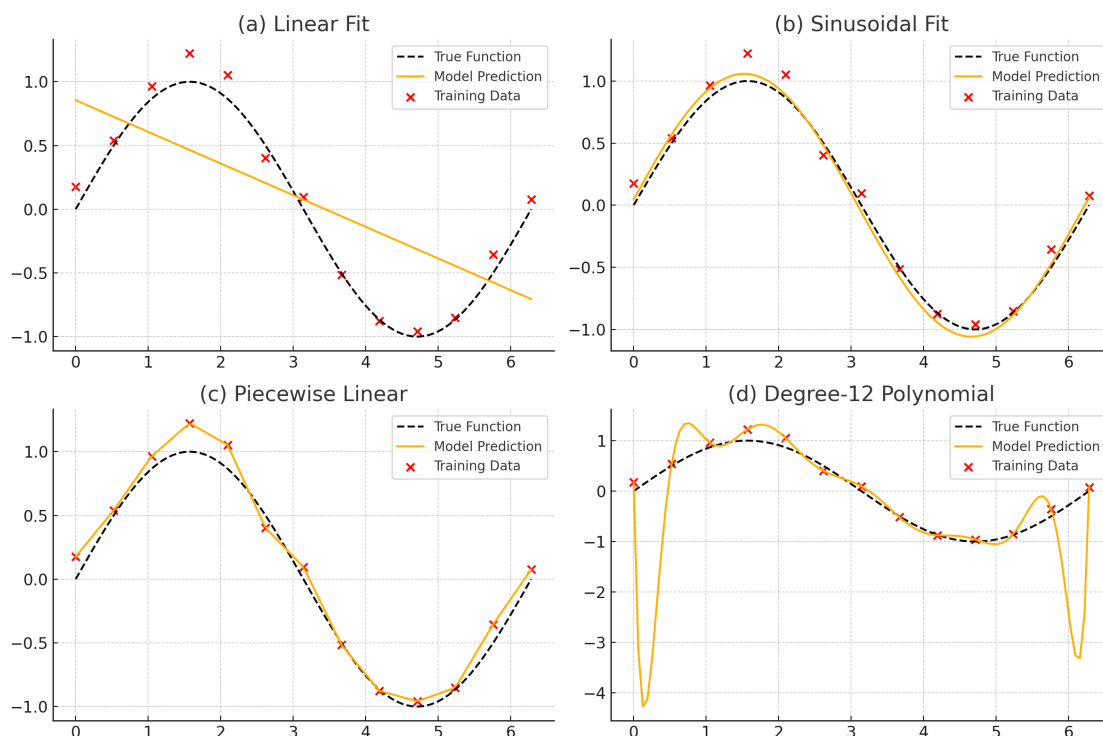


Figure 1.4: Bias-Variance Tradeoff Visualization

What is a Good Fit?

A good fit in machine learning is a model that captures the underlying structure of the data without memorizing the noise. It achieves a balance between underfitting and overfitting.

Characteristics of a Good Fit

- **Balanced bias and variance:** The model is neither too simple nor too complex.
- **Low generalization error:** It performs well on both the training set and unseen test data.
- **Appropriate complexity:** The model captures meaningful trends without reacting to random fluctuations.

Example

- A good fit curve for noisy sine-wave data 1.4 would follow the overall shape of the sine wave (subplot b) while ignoring random noise in individual data points.

A good fit is one that generalizes well — it learns the signal, not the noise.